International Cybersecurity and Digital Forensics Academy (ICDFA)

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Registration Number	C11/26/PCDFIR
Course Code	CIP-B101
Course Title	Basic Computer Skills for Digital Forensic
Lab Title	PC System Fundamentals, Disk Sectors and Partition Analysis
Lab Code	Lab1B
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Learning Outcomes

- Explain how computer system layers generate digital evidence.
- Identify basic PC hardware, operating system, storage, memory, and processor information.
- Differentiate between a program, process, and thread in a forensic context.
- Explain why computer forensics is technically difficult due to operating systems, applications, hardware, connected devices, and data volume.
- Describe disk sectors and calculate disk or partition sizes using sector count and sector size.
- Use safe read-only commands to view disk and partition information.
- Explain the PC boot process from power-on to operating system control.
- Compare MBR and GPT partitioning concepts at beginner level.

Tools Used

- Windows 11Pro System
- PowerShell
- Kali-Linux
- WinHex
- CMD
- DiskPart
- Snipping tool

A1. Capture basic system specification

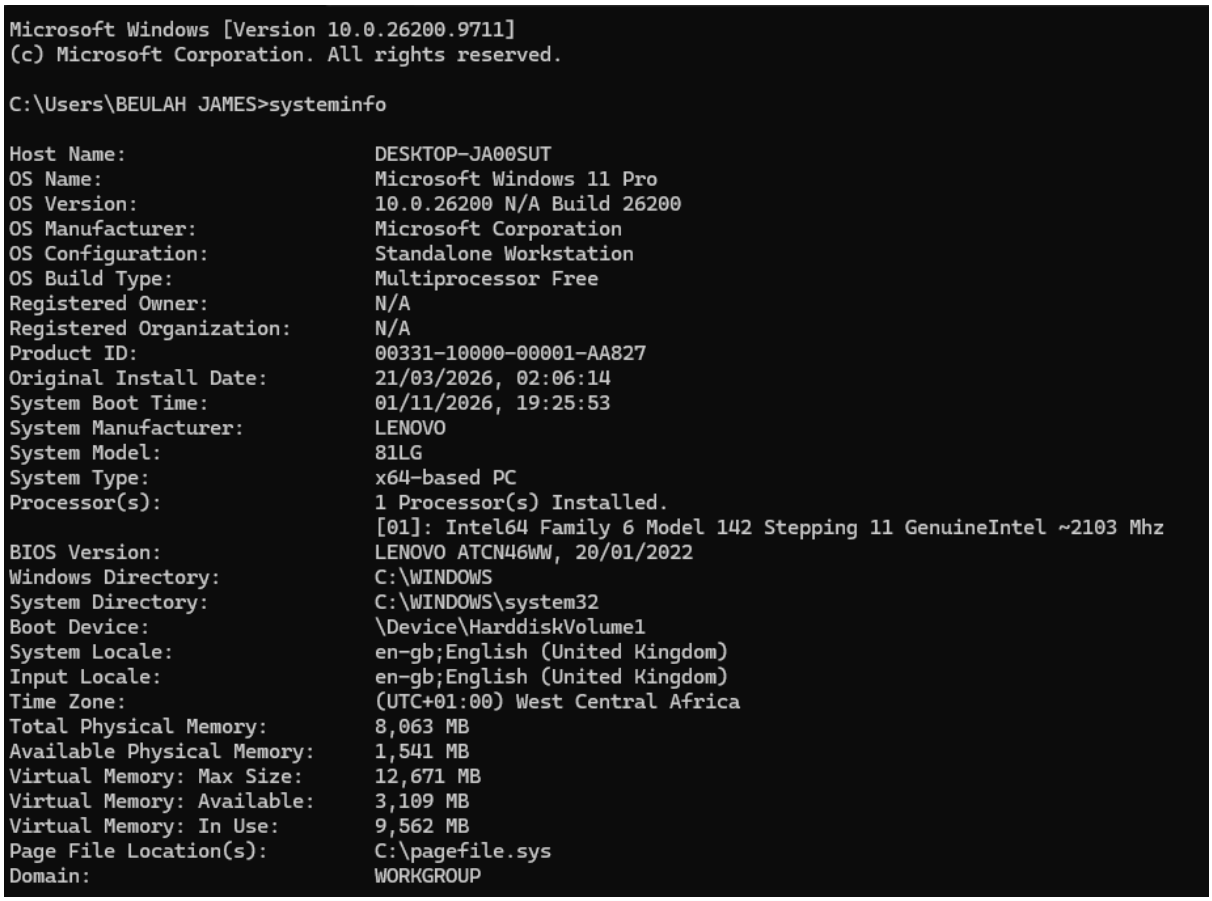
I opened the system information by pressing windows + R, then I typed msinfo32, and pressed Enter, a window opened that displayed the basic system specification as seen in FigA1 and FigA1b below, while FigA1c shows the systeminfo command typed on the windows command prompt.

Item	Value
OS Name	Microsoft Windows 11 Pro
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	DESKTOP-JA00SUT
System Manufacturer	LENOVO
System Model	81LG
System Type	x64-based PC
System SKU	LENOVO_MT_81LG_BU_idea_FM_IdeaPad L340-15IWL
Processor	Intel(R) Core(TM) i3-8145U CPU @ 2.10GHz, 2304 Mhz, 2 Core(s), 4 Logical Pr...
BIOS Version/Date	LENOVO ATCN46WW, 20/01/2022
SMBIOS Version	3.0
Embedded Controller Version	1.46
BIOS Mode	UEFI
BaseBoard Manufacturer	LENOVO
BaseBoard Product	LNVNB161216
BaseBoard Version	SDK0J40700 WIN
Platform Role	Mobile
Secure Boot State	Off
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32

FigA1 Screenshot Showing Basic System Specification

Item	Value
BaseBoard Version	SDK0J40700 WIN
Platform Role	Mobile
Secure Boot State	Off
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32
Boot Device	\Device\HarddiskVolume1
Locale	United Kingdom
Hardware Abstraction Layer	Version = "10.0.26100.1"
Username	DESKTOP-JA00SUT\BEULAH JAMES
Time Zone	W. Central Africa Standard Time
Installed Physical Memory (RAM)	8.00 GB
Total Physical Memory	7.87 GB
Available Physical Memory	1.76 GB
Total Virtual Memory	12.4 GB
Available Virtual Memory	3.46 GB
Page File Space	4.50 GB
Page File	C:\pagefile.sys
Kernel DMA Protection	Off
Virtualisation-based security	Running
Virtualisation-based security re...	
Virtualisation-based security av...	Base Virtualisation Support, DMA Protection, UEFI Code Readonly, SMM Secu...

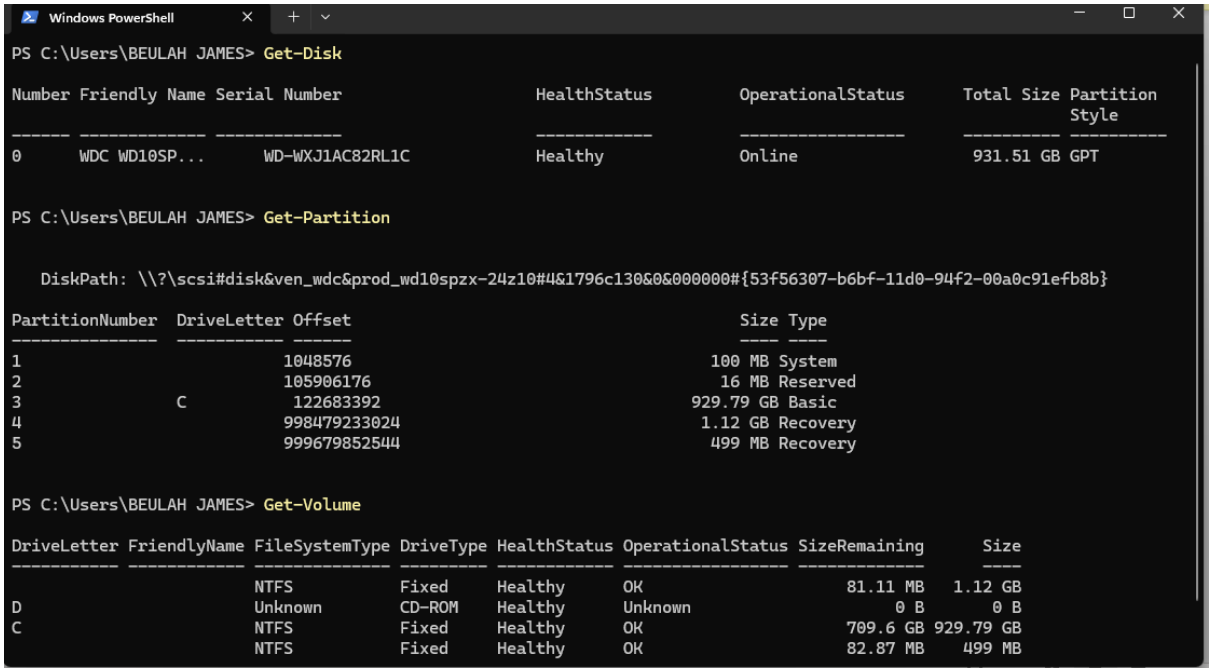
FigA1b Screenshot Showing Installed Physical memory



FigA1c Screenshot Showing First Page of Systeminfo on cmd

A2. Capture storage information

I opened PowerShell by pressing windows + R then I typed PowerShell, the terminal opened then I run the commands that were given. The first two commands didn't then I proceeded to use the PowerShell alternative commands, that worked then I captured the screenshots of storage model, size, partitions, and file systems as seen in FigA2 below.



FigA2 Screenshot Showing Storage Information

A3 Map evidence sources to system layers

Layers/Source	Type of Evidence	Where It May Be Found	Why it Matters
User Layer	User login and logout records, user account creation and deletion, file creation, password change records	Authentication logs, directory services, file system audit log, application event log, password policy	They provide verifiable, time-stamped trails of user activity
Application Layer	Email communication record, download records, document access and editing, browser cookies and cached files.	Application logs, mail servers, browser database record, application caches.	Shows user activity and intent, establish timelines and verify communication.
System Call (OS) Layer	File access and modification, memory allocation activities, process creation request.	Process monitor traces, syscall log, windows security auditing.	Shows unusual file access, unexpected memory allocation.
Kernel(OS) Layer	System crashes, process and thread activity, bug checks, memory management.	Memory dumps, crash dumps, windows event log.	Shows unauthorized activity, it identifies root cause of crashes.

Device Driver	Driver installation records, driver failures, device connection and disconnection error.	Device manager information, windows system event logs, registry.	Shows activity of physical devices.

TableA3 Mapping of Evidence Source to System Layers

Explanation

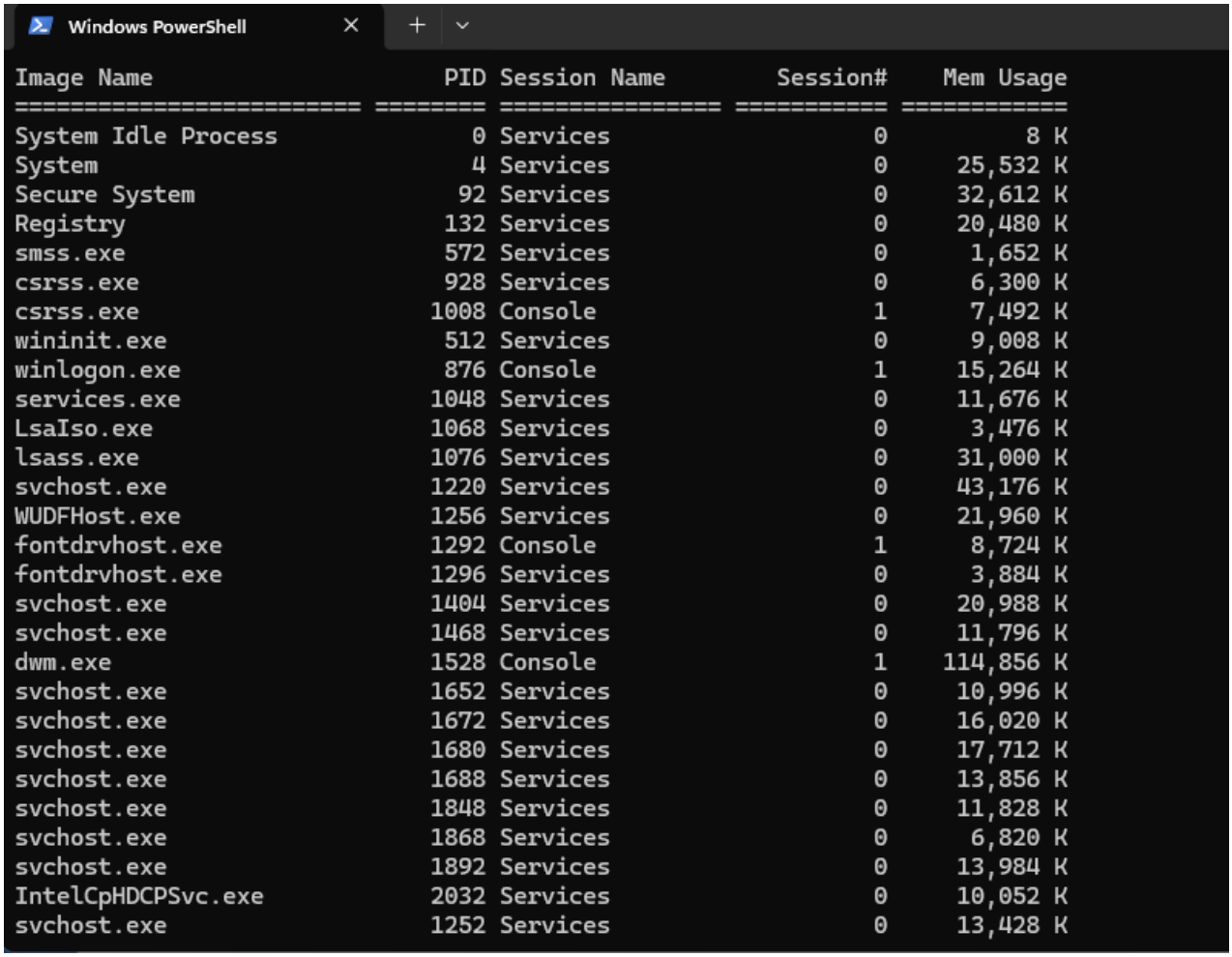
Evidence may exist in more than one layer because the system layers work together with each layer performing its specific role on the same operation, and also generates, processes, stores, or transfers information. For example, when I downloaded my course modules from resources on the ICDFA e-campus portal, I instructed the computer to download the file, the browser processed my request, then the application requested the operating system services, which manages the download process and determines where the file will be saved and the file system creates the file entry and records the information, the downloaded is then saved on my hard drive.

This activity leaves traces on the browser, the operating system, file system and storage device.

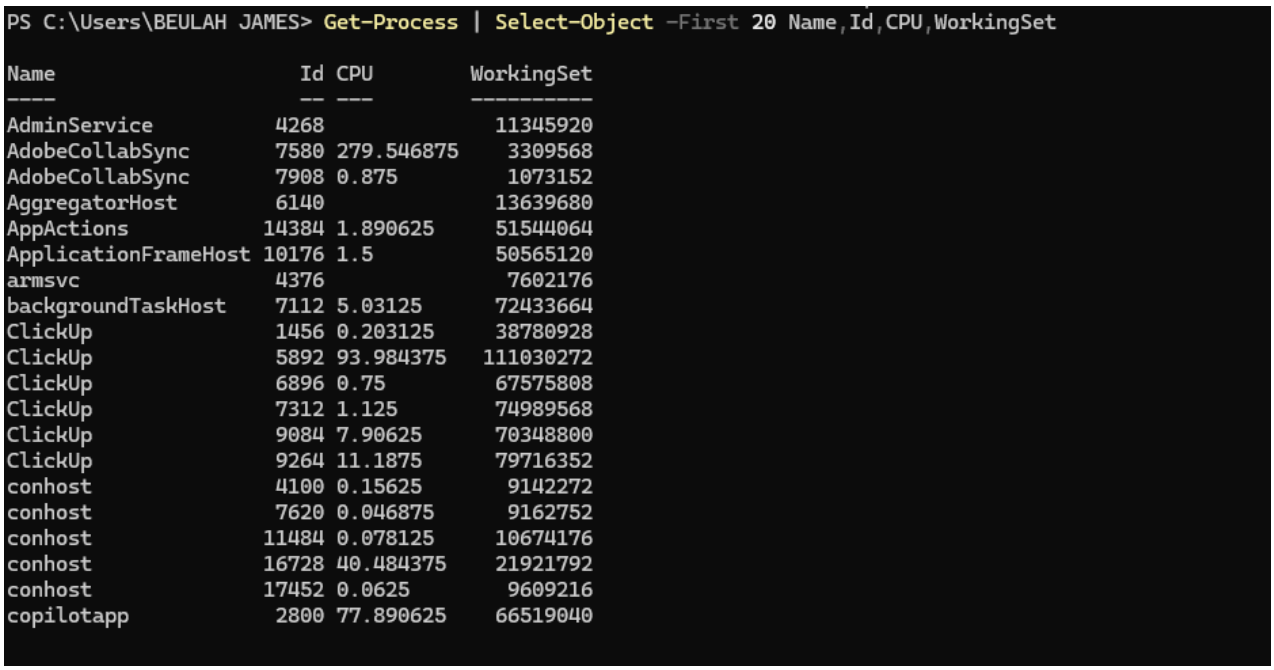
B1. Observe running processes

I opened PowerShell by pressing windows + R then I typed PowerShell, the terminal opened then I run the commands that were given, I tried both commands because I wanted to see if it would return the same output, the screenshots are FigB1 and FigB1b below.

I selected three process records from FigB1 and recorded their process names, PID values, and likely purpose as seen in TableB1.



FigB1 Screenshot Showing Running Processes



FigB1b Screenshot Showing 20 Process Name ID CPU

S/N	Process Name	Process ID	Purpose
1	System	4	Manages core windows operating system functions, handles hardware and device-driver operations, coordinates certain memory and processor activities, it also supports communication between hardware and software.
2	Secure System	92	Supporting credentials guard, protecting sensitive operating-system security processes, supporting memory integrity.
3	Registry	132	Hardware configuration, software setting, user preferences, system startup and service configurations

TableB1 Showing Process Names, PID Value and Purpose

B2. Explain the Difference

Program	Process	Thread
A program is a set of instructions usually codes written to perform a task, the codes are stored in a memory and is passive.	A process is a program currently running within an operating system, it usually active and has a process identification value and memory allotted to it.	A thread is a smallest unit running within a process that enables a program to perform multiple tasks concurrently while sharing the same memory and resources.

TableB2 Difference Between a Program, Process and Thread

Process information is very useful during malware analysis and live forensic triage because it helps investigators understand what programs are currently running on a computer, what they are doing, and whether any of them are behaving suspiciously. It helps investigators identify programs that may be malicious and to understand what the malware is doing after it enters a computer.

C1 Reproduce the Slide Calculation

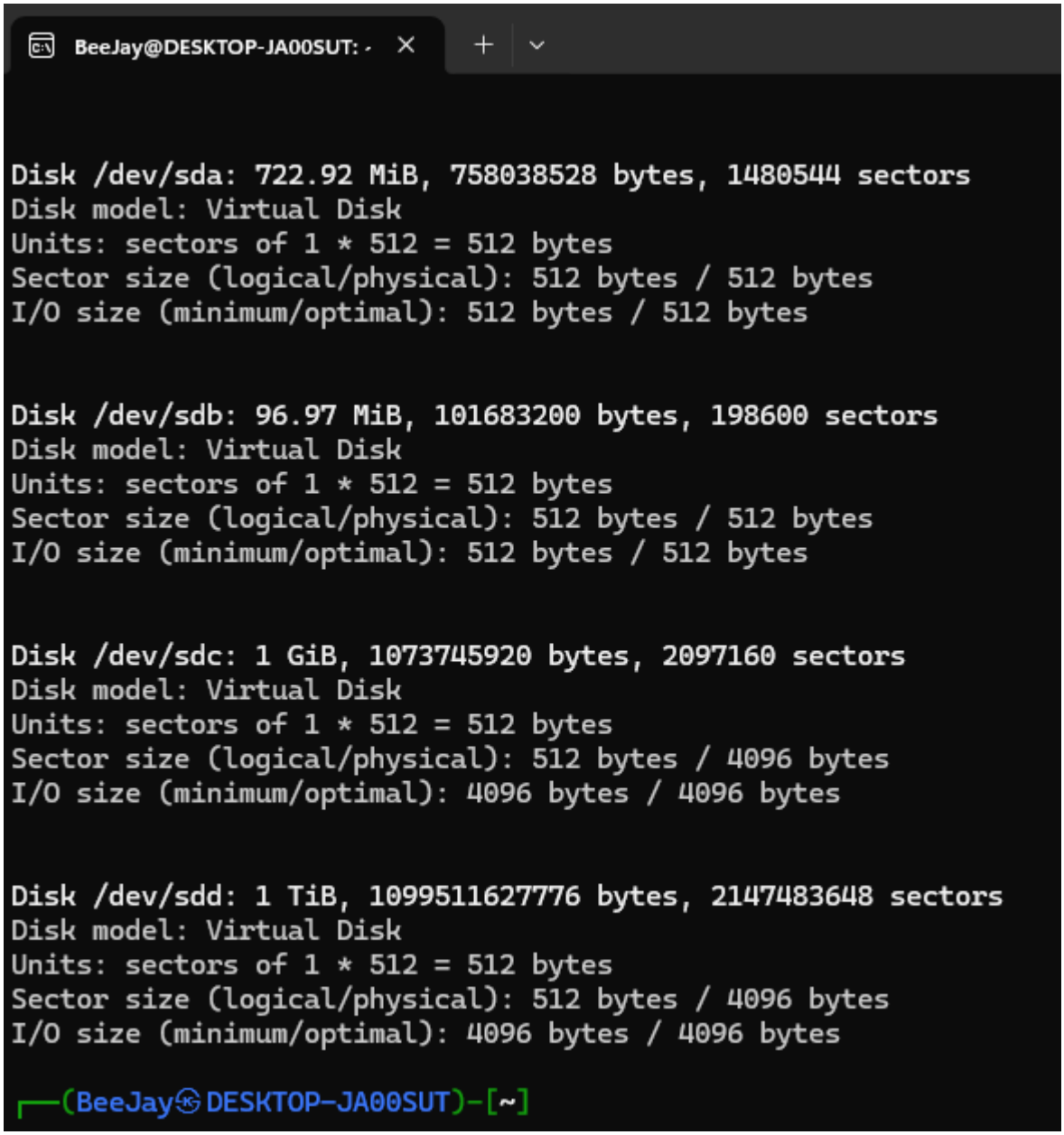
sectors per track = 400, heads = 12, cylinders = 17000, and bytes per sector = 512.
Total sectors = sectors per track * no. of heads * cylinders
Total disk size = total sectors * bytes per sector
1024bytes =1KB 1024KB = 1MB 1024MB = 1GB
To get values in KB, MB and GB divide the respective values by 1024

Item	Formula/Value	Answer
Total sectors	400 * 12 * 17000	81,600,000 sectors
Total disk size in bytes	81,600,000 * 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

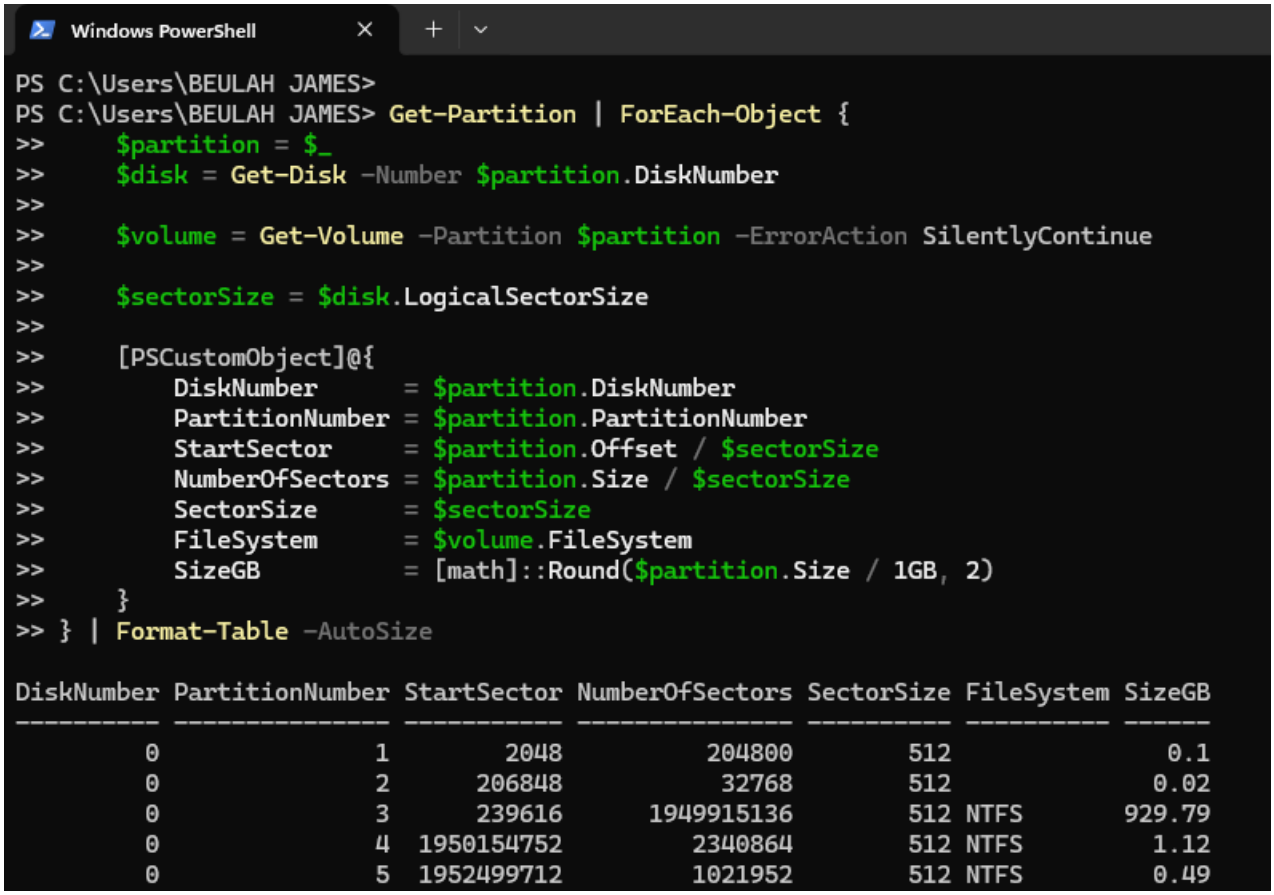
TableC1 Slide Calculations

C2 Calculate Your Own Disk or Partition Size

The screenshots in FigC2 and FigC2b shows the disk details and partition table from Kali-Linux and PoweShell respectively. I first tried the code ‘sudo fdisk -l the output did not display the start sector and the filesystem so I used an alternative code on PowerShell to get the partition table that outputs all required details. TableC2 shows the two partitions identified from FigC2b and the required details.



FigC2 Screenshot Showing Disk Details Using Kali-Linux



FigC2b Screenshot Showing Partition Table Using PowerShell

Partition	Start Sector	No. of Sectors	Sector size	File system
3	239,616	1,949,915,136	512	NTFS
4	1,950,154,752	2,340,864	512	NTFS

TableC2 Partition Table

From my partition table in TableC2
Partition 3 no of sectors = 1,949,915,136 sector size = 512
Total disk size = total sectors * sector size
1024bytes =1KB 1024KB = 1MB 1024MB = 1GB
To get values in KB, MB and GB divide the respective values by 1024

Item	Formula/Value	Answer
Total disk size in bytes	1,949,915,136 * 512	998,356,549,632 bytes
Kilobytes	998,356,549,632 / 1024	974,957,568 KB
Megabytes	974,957,568 / 1024	952,107 MB
Gigabytes	952,107/ 1024	929.79 GB

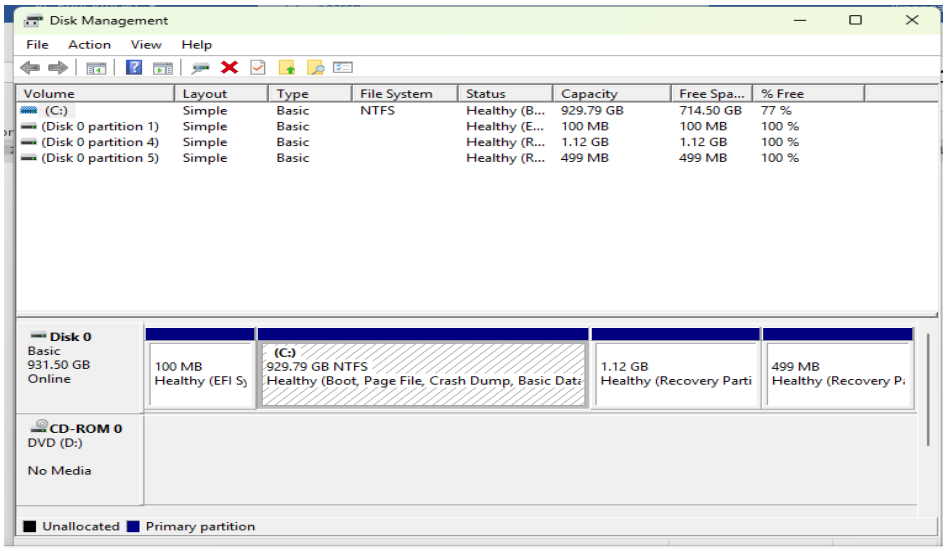
TableC2b Showing Partition 3 Disk Size Step-by-Step Calculation

From my partition table in TableC2
Partition 4 no of sectors = 2,340,864 sector size = 512
Total disk size = total sectors * sector size
1024bytes =1KB 1024KB = 1MB 1024MB = 1GB
To get values in KB, MB and GB divide the respective values by 1024

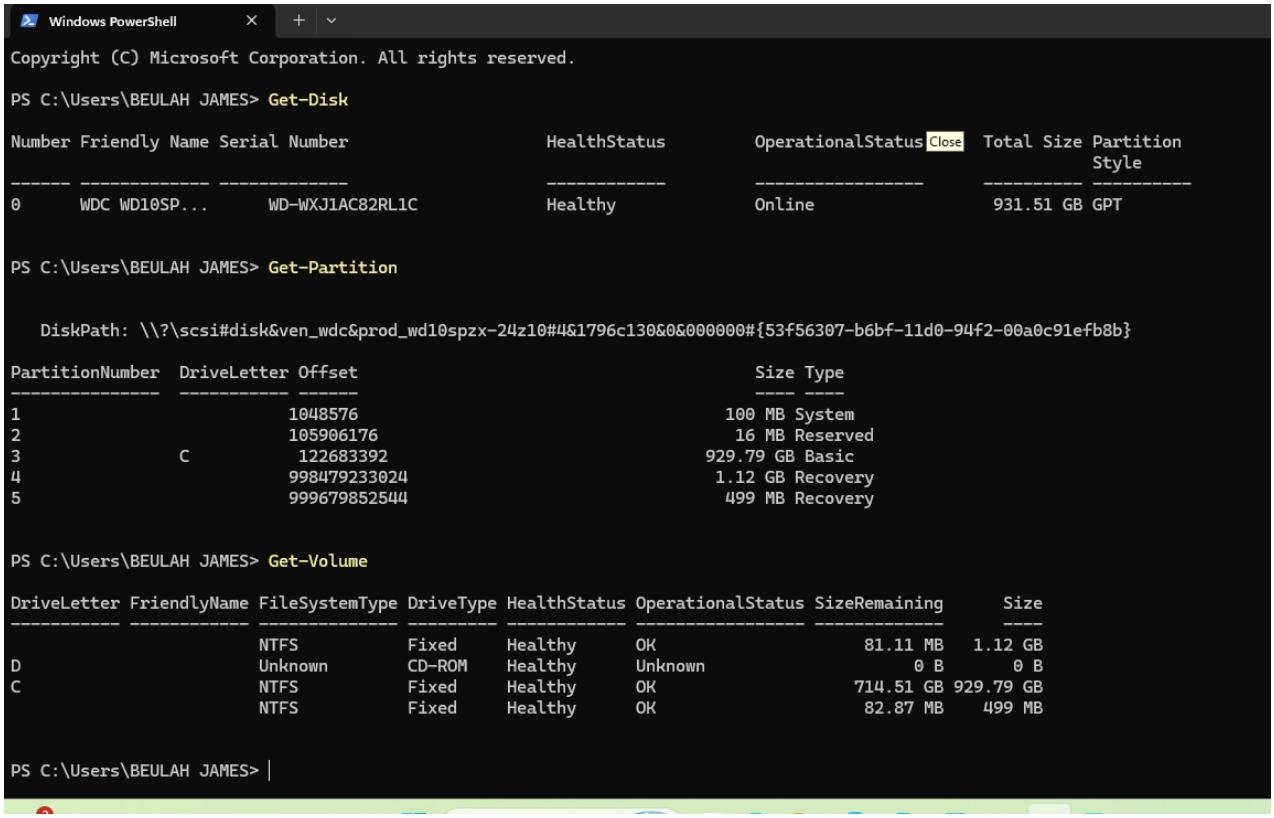
Item	Formula/Value	Answer
Total disk size in bytes	2,340,864 * 512	1,198,522,368 bytes
Kilobytes	1,198,522,368 / 1024	1,170,432 KB
Megabytes	1,170,432 / 1024	1,143 MB
Gigabytes	1,143 / 1024	1.12 GB

TableC2b Showing Partition 4 Disk Size Step-by-Step Calculation

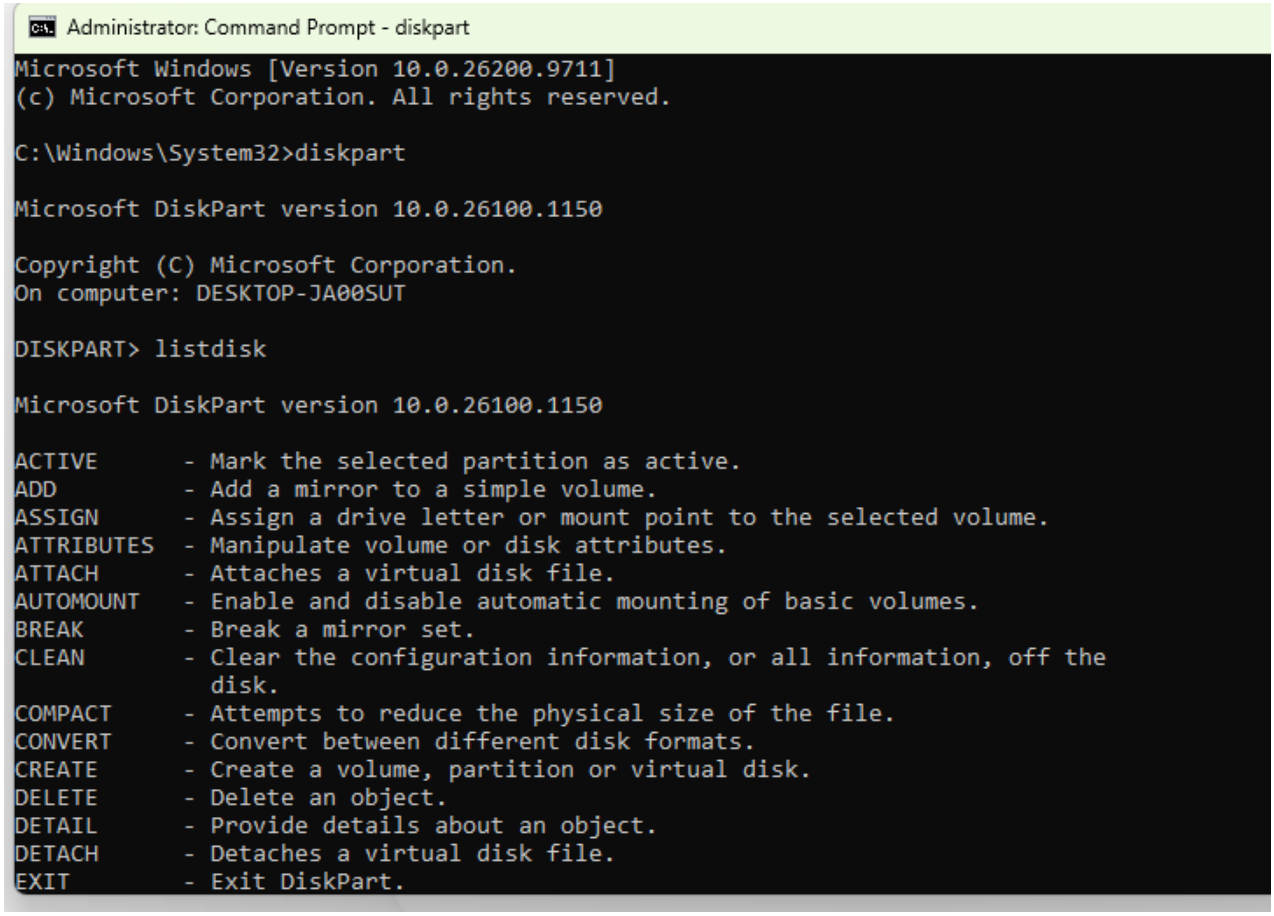
D1. Windows method - Disk Management, PowerShell and DiskPart



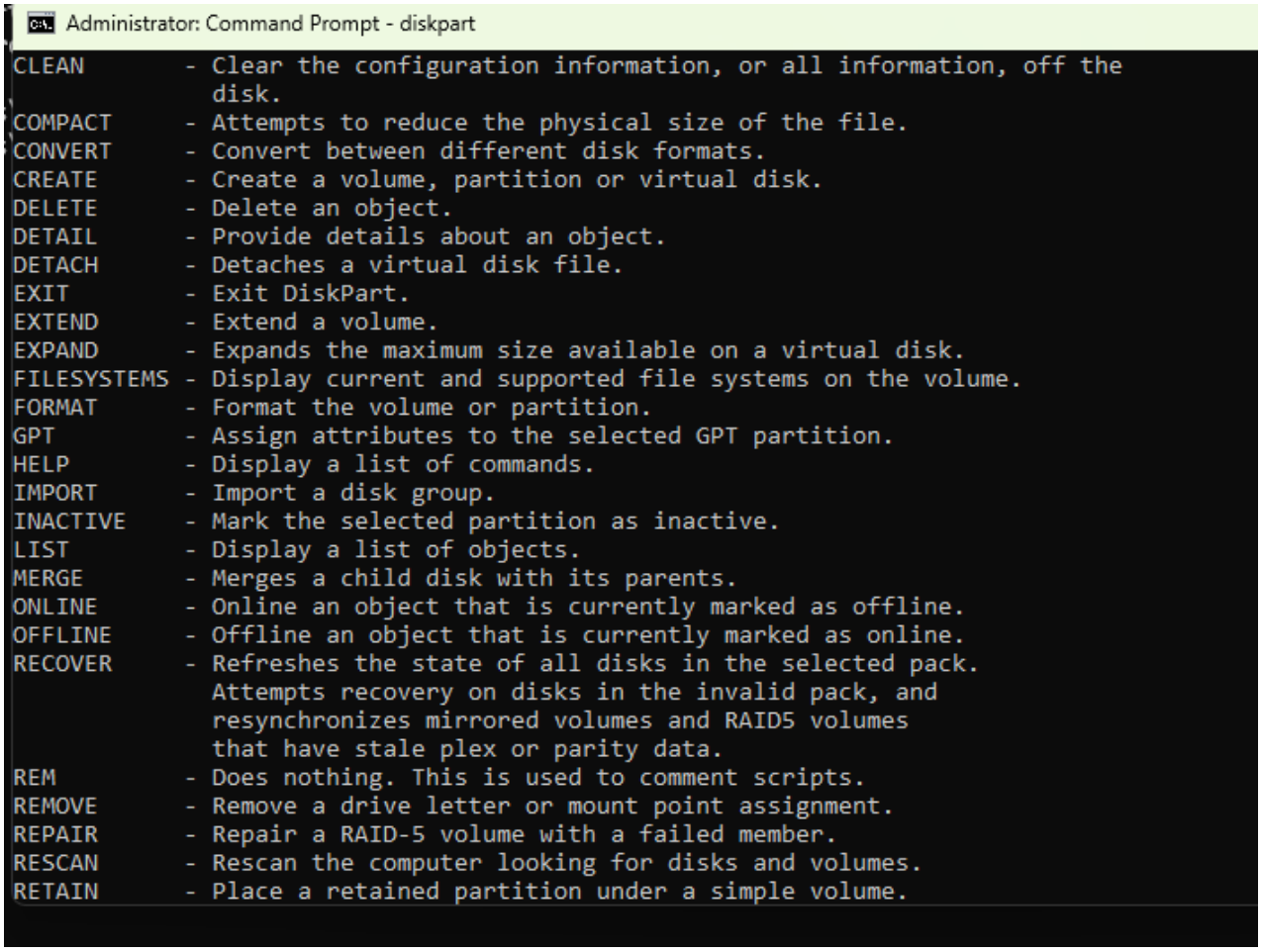
FigD1 Showing Disk Management



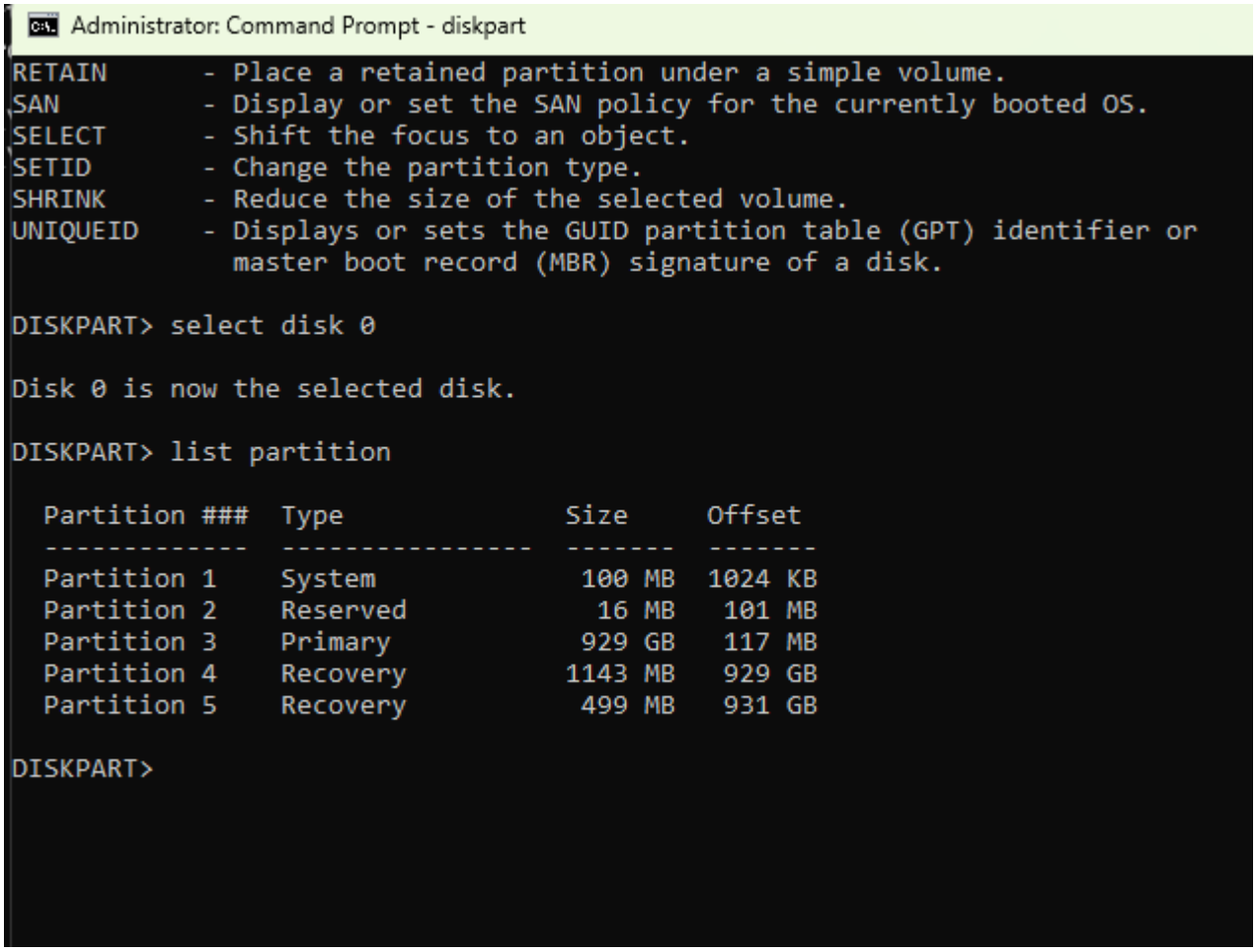
FigD1b Screenshot Showing Disk Partition and Volume Using PowerShell



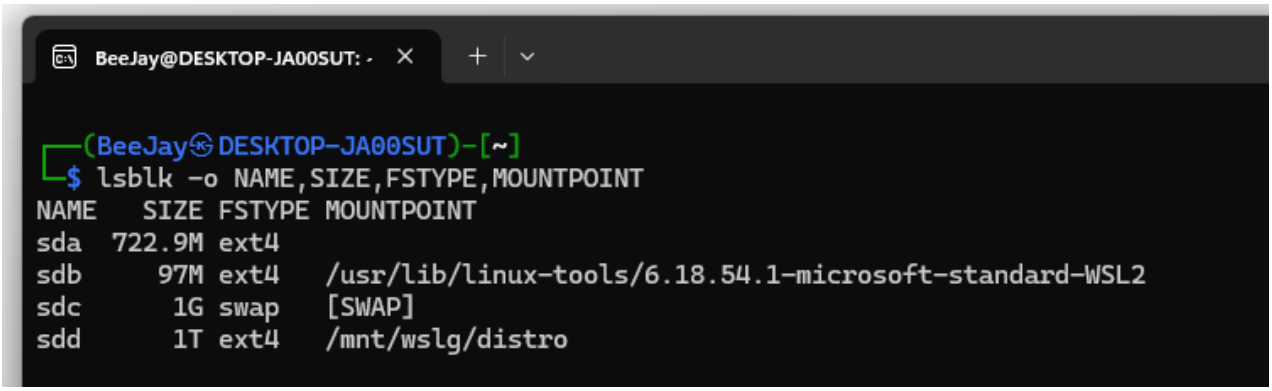
FigD1c Screenshot Showing Diskpart Using Adminstrator CommandPrompt



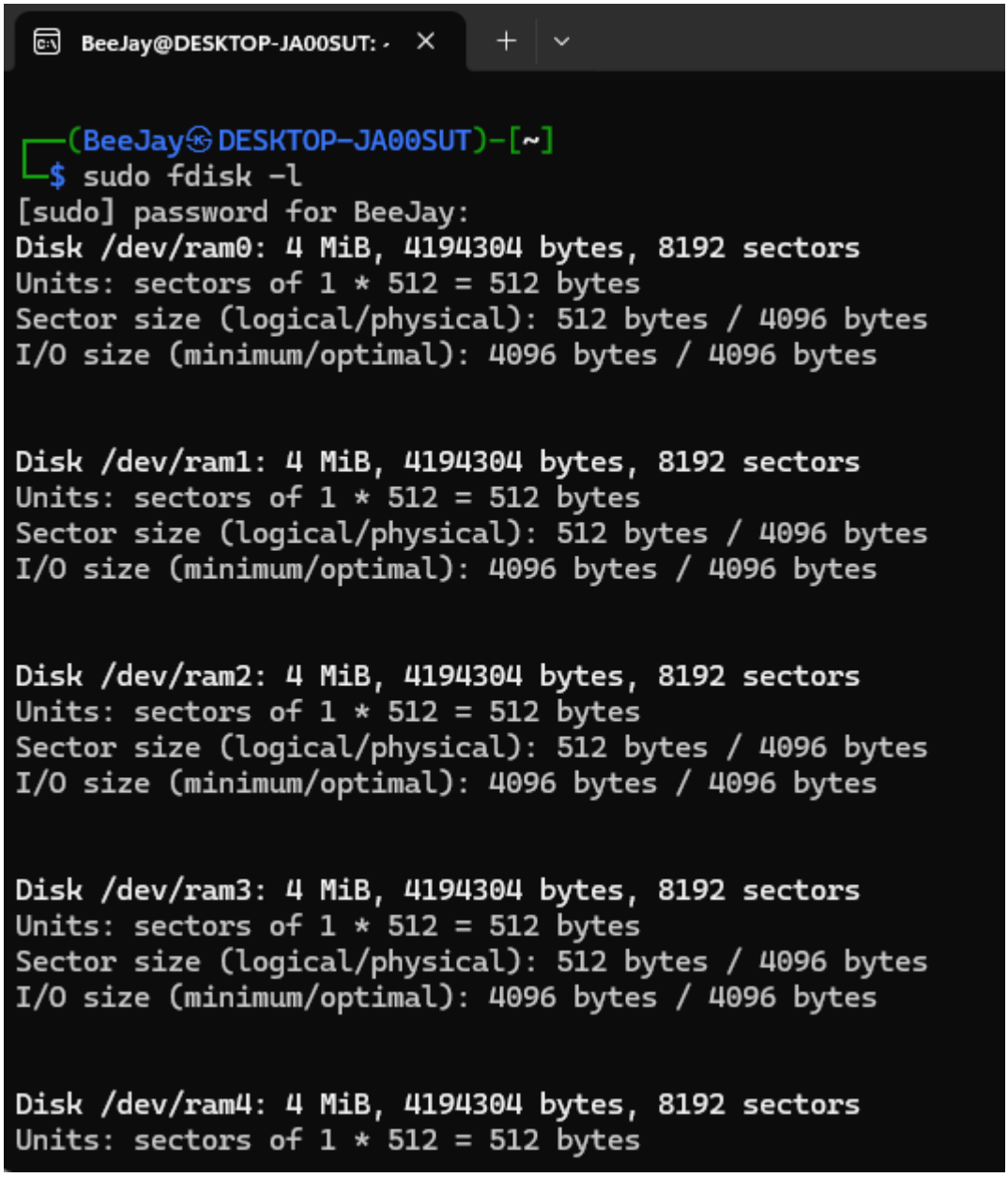
FigD1d Screenshot Diskpart Details Using Administrator CommandPrompt



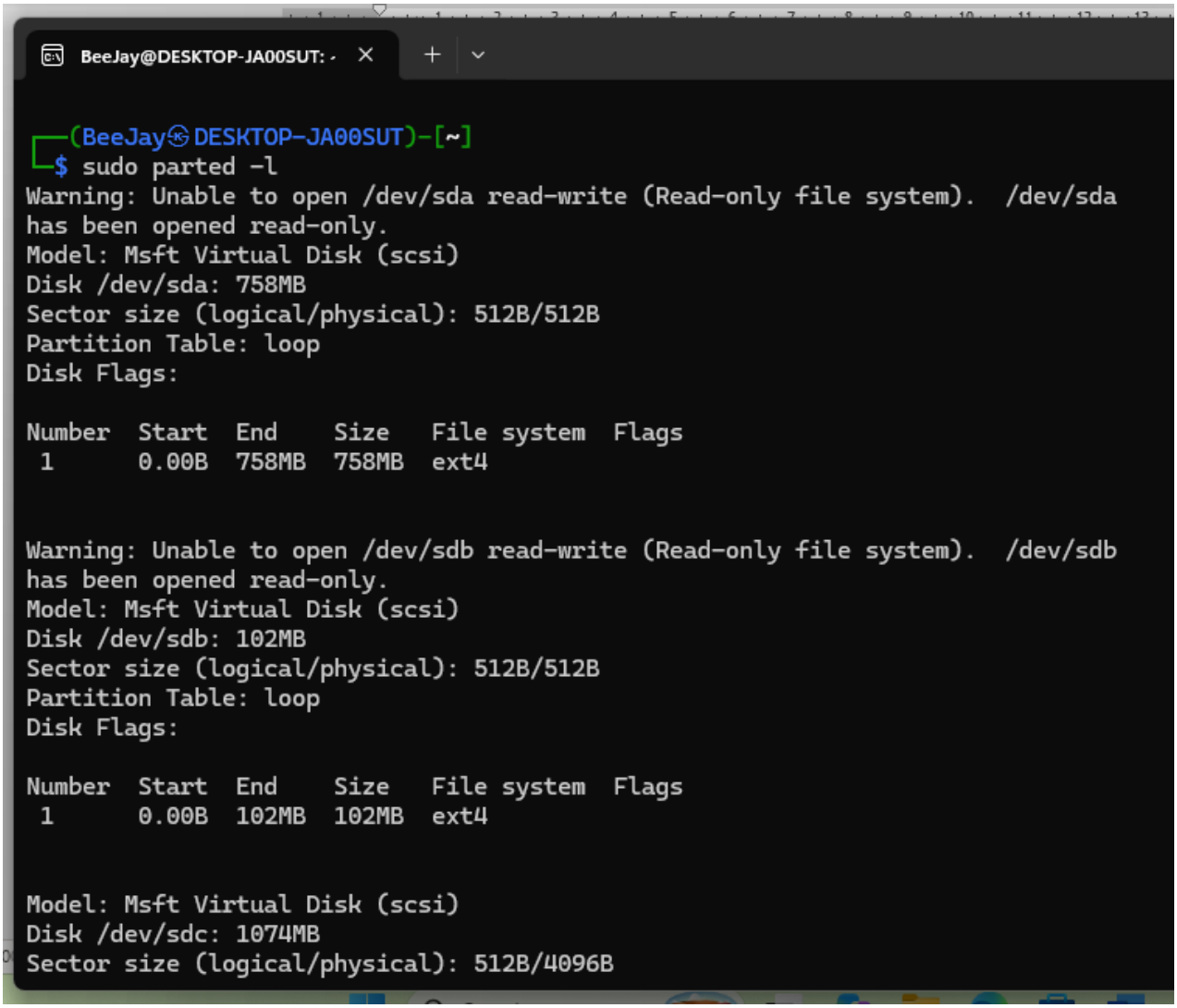
FigD1e Screenshot Showing List Partition Using Administrator CommandPrompt



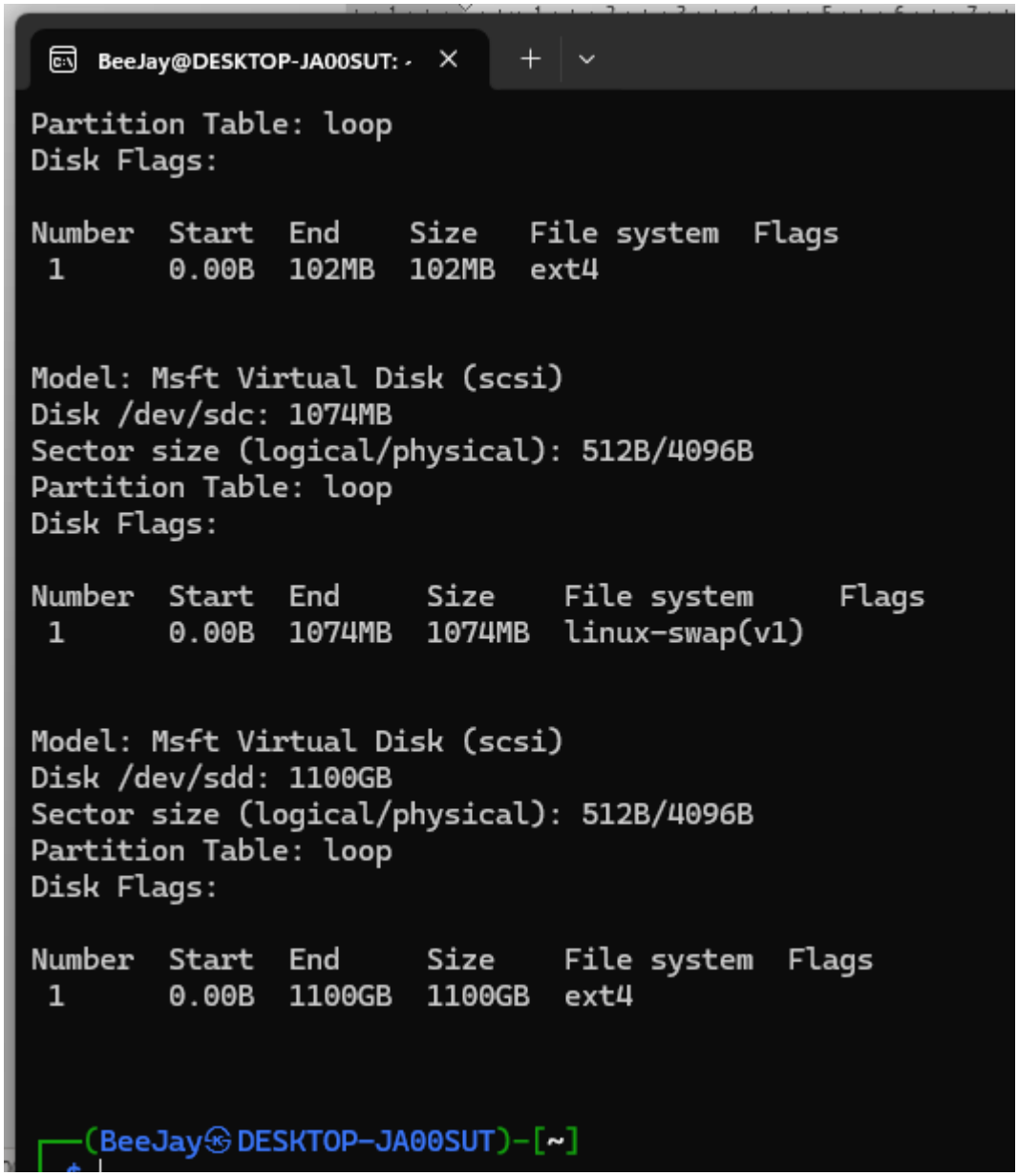
FigD2 Screenshot Showing KaliLinux Method



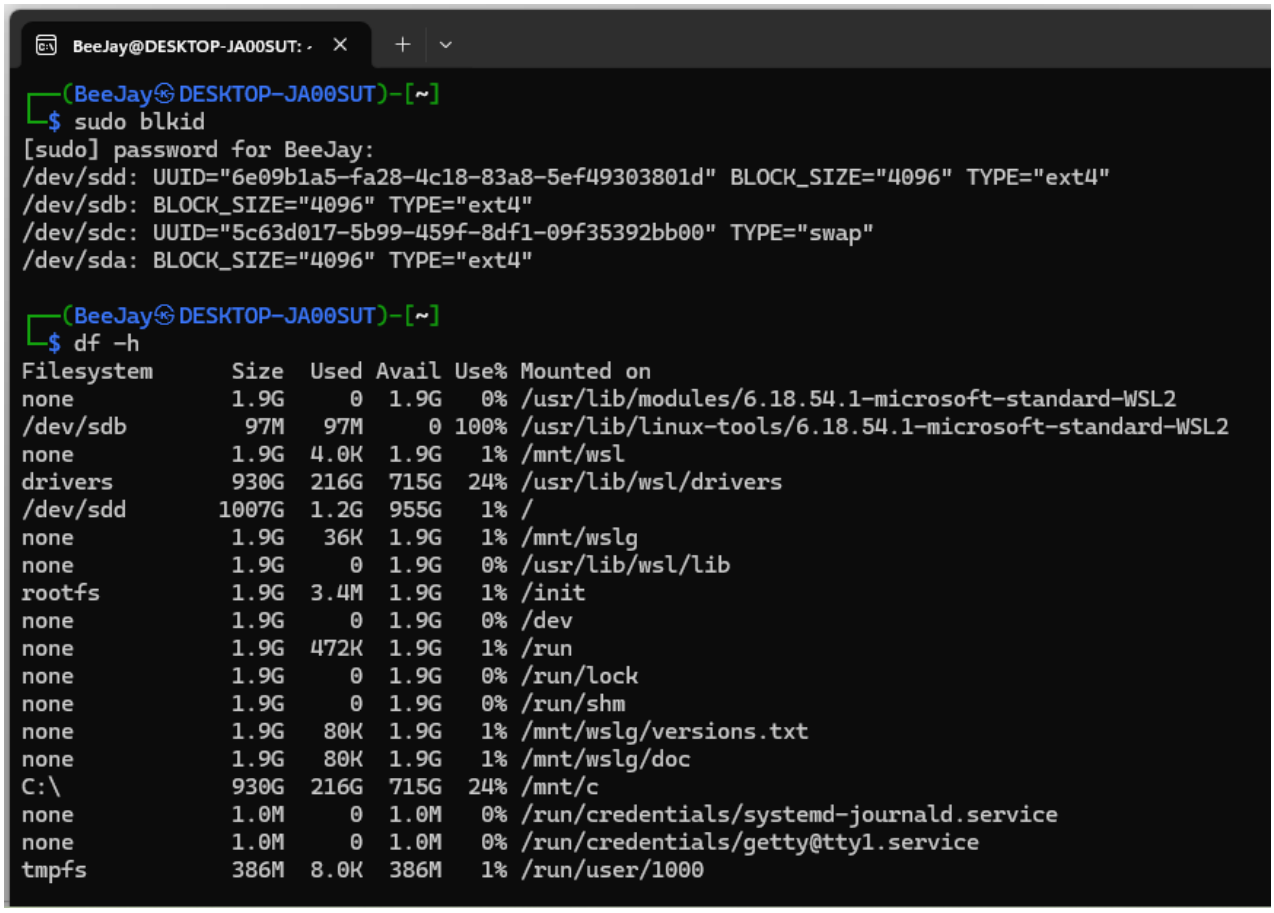
FigD2b Screenshot Showing Disk details with KaliLinux



FigD2c Screenshot Showing Partition details with KaliLinux



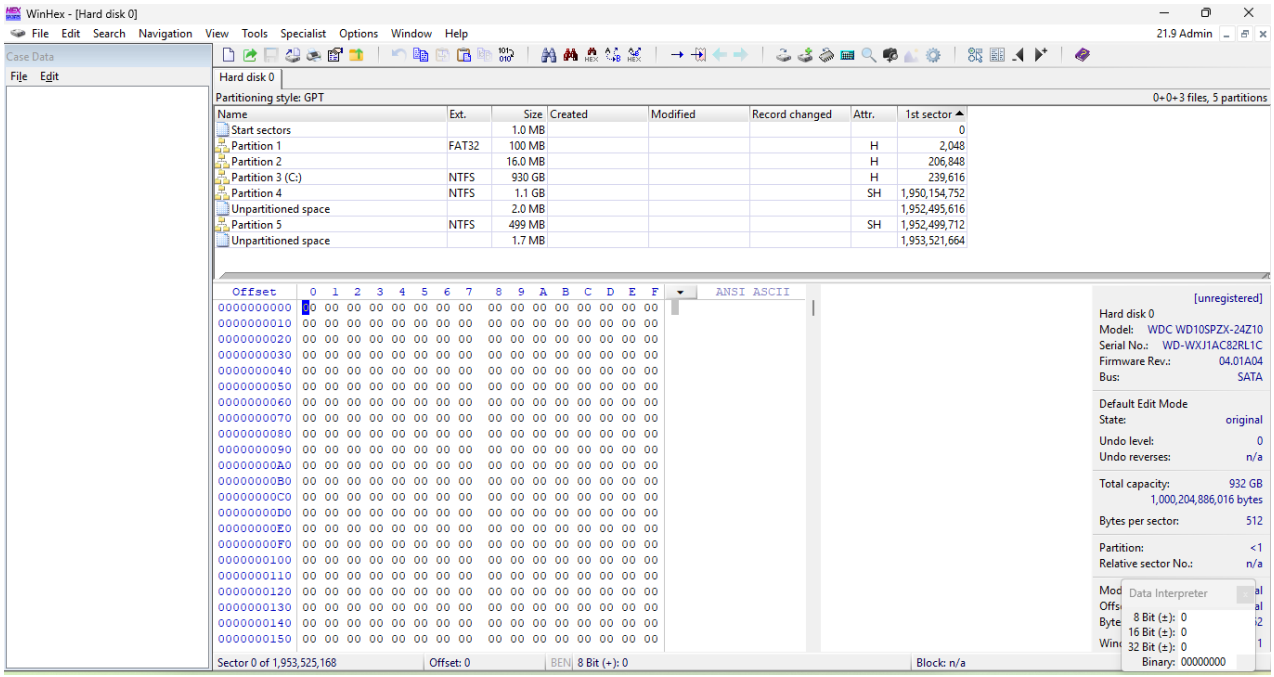
FigD2d Screenshot Showing Partition Size and Filesystem details with KaliLinux



FigD2e Screenshot Showing BlockID and Diskspace on KaliLinux

D3. Optional WinHex or Forensic viewer method

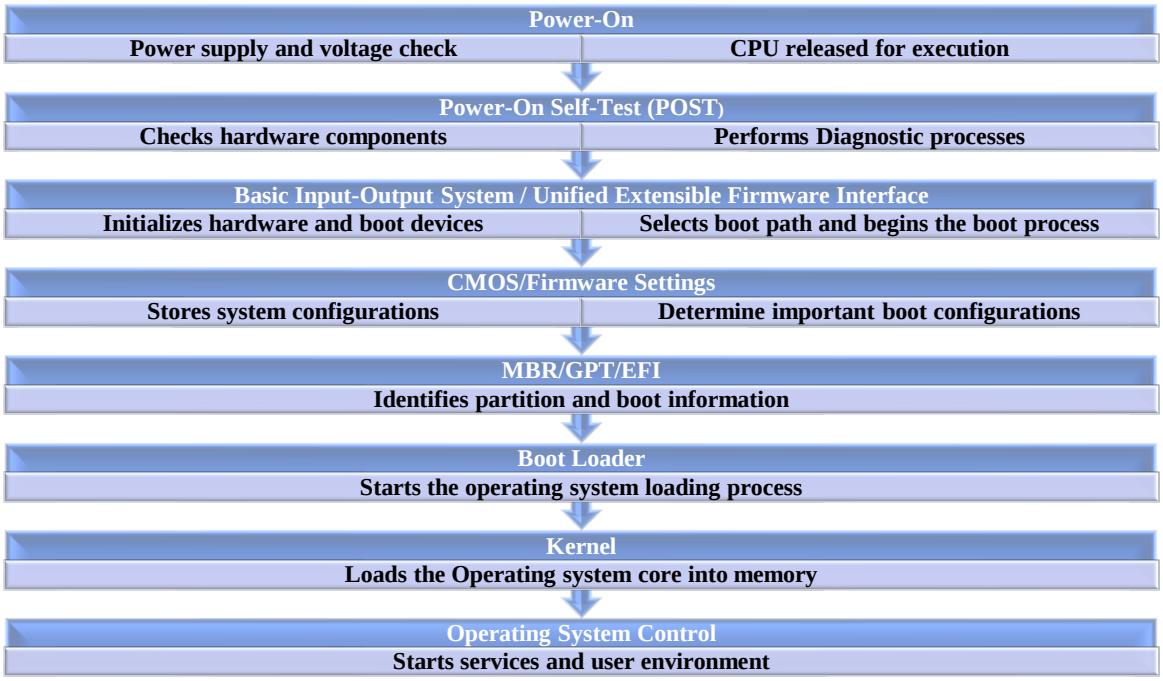
In this screenshot it is identified the disk is using GPT



FigD3 Screenshot Showing Details Using WinHex

E1 Explain the Boot Process

A Flow Process Explaining How a PC boots.



My system Boots from UEFI based on system information evidence as seen below

Item	Value
OS Name	Microsoft Windows 11 Pro
Version	10.0.26200 Build 26200
Other OS Description	Not Available
OS Manufacturer	Microsoft Corporation
System Name	DESKTOP-JA00SUT
System Manufacturer	LENOVO
System Model	81LG
System Type	x64-based PC
System SKU	LENOVO_MT_81LG_BU_idea_FM_IdeaPad L340-15IWL
Processor	Intel(R) Core(TM) i3-8145U CPU @ 2.10GHz, 2304 Mhz, 2 Core(s), 4 Logical Pr...
BIOS Version/Date	LENOVO ATCN46WW, 20/01/2022
SMBIOS Version	3.0
Embedded Controller Version	1.46
BIOS Mode	UEFI
BaseBoard Manufacturer	LENOVO
BaseBoard Product	LNVNB161216
BaseBoard Version	SDK0J40700 WIN
Platform Role	Mobile
Secure Boot State	Off
PCR7 Configuration	Elevation Required to View
Windows Directory	C:\WINDOWS
System Directory	C:\WINDOWS\system32

Stage	What Happens	Forensic Relevance
Power-On	Supplies power and starts CPU	Power event investigation
POST	Performs diagnostic checks on hardware and reports serious problem when detected	Hardware failure evidence
BIOS /UEFI	Initializes hardware and boot devices, selects boot path and begins the boot process	Firmware configuration
CMOS/Firmware settings	Stores system configuration and determines important boot configurations	Boot order and time settings
MBR/GPT/EFI	Identifies partition and boot information	Partition analysis
Boot Loader	Starts operating system loading process	Boot configuration and tampering
Kernel	Loads the operating system core into memory	Memory and rootkit analysis
Operating System control	Starts services and user environment	Logs, processes and user activity

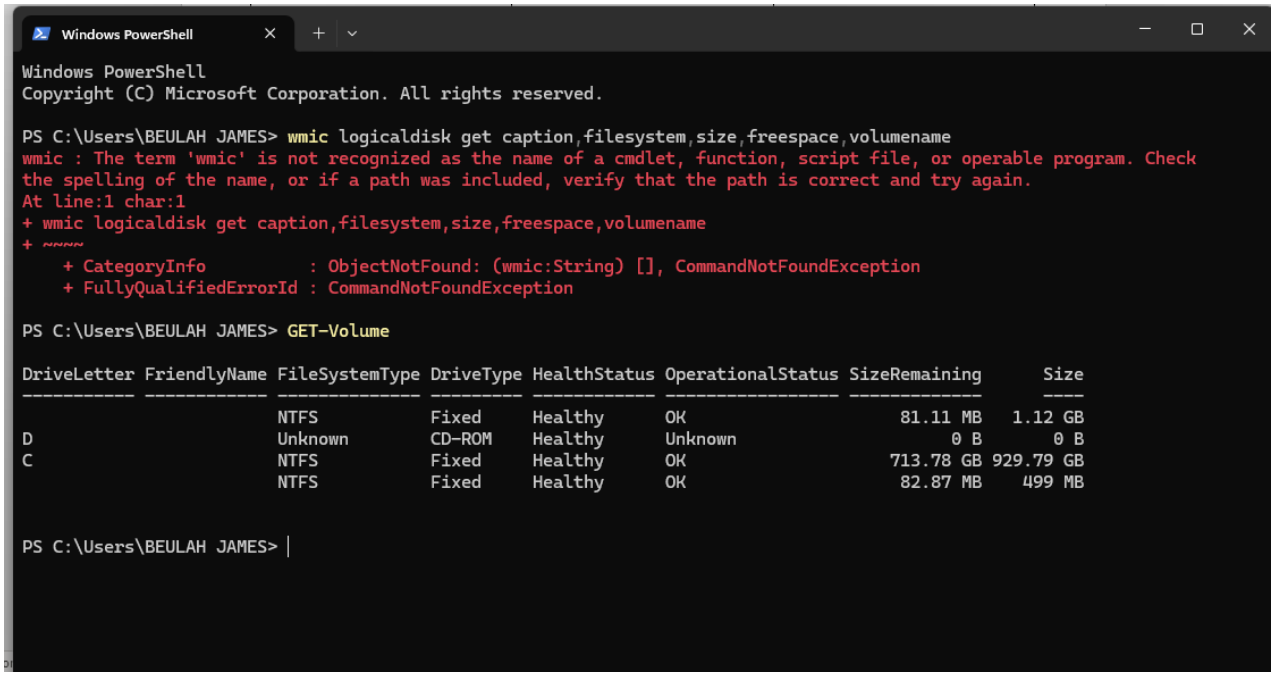
E2. Compare MBR and GPT

Feature	MBR	GPT
Full meaning	Master Boot Record	GUID Partition Table
Age	Older	Modern
Common firmware association	Legacy BIOS	UEFI
Maximum disk size	Approximately 2 TB	Much larger
Primary partitions	Maximum 4	Typically, 128 in Windows
Partition identification	Partition type codes	Unique GUIDs
Backup partition information	No standard backup	Yes
Data integrity checks	No equivalent GPT CRC structure	CRC checksums
Boot information	Boot code in MBR	UEFI loads EFI boot files from ESP
Reliability	More limited redundancy	More robust partition metadata

TableE2 Comparison Between MBR and GPT

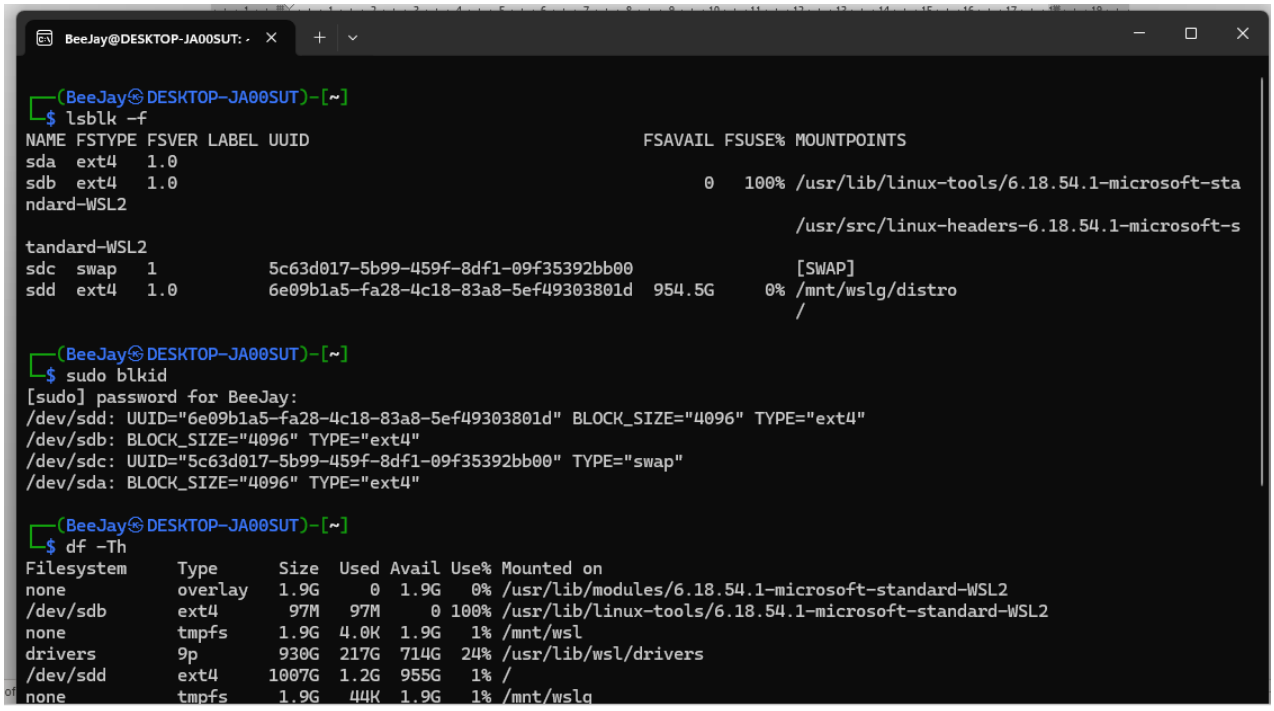
F1 Part F - File System Identification and Interpretation

The file system used by two Volumes on my system is NTFS as shown in FigF below



FigF Screenshot Showing File System

In Linux partitions use both ext4 and swap as shown in FigF1 below



FigF1 Screenshot Showing Linux File System

A file system is the method an operating system uses to organise, store, name, manage and retrieve data on a storage device such as an HDD, SSD, USB flash drive or memory card. These data are organized as files into folders and directories, so in digital forensics, a file system is

important because it helps investigators locate, examine, recover and interpret digital evidence stored on a computer. Investigators can locate documents, photographs, videos, emails and other files that may contain evidence.

Part G

Task	Required Evidence	Student Notes / Answer
Identify partition 1	Screenshot of partition table and command output	Partition table and command output is seen in FigC2b
Record partition 1 sector count	Start sector, end sector or sector count	Start sector = 2048
Calculate partition 1 size	Formula and result in GB	0.1GB as shown in TableG below
Identify partition 2	Screenshot of partition table and command output	Partition table and command output is seen in FigC2b
Record partition 2 sector count	Start sector, end sector or sector count	Start sector = 206848
Calculate partition 2 size	Formula and result in GB	0.02GB as shown in TableG1 below
Compare with OS-reported size	Disk Management, Get-Volume, lsblk, or df output	The results are the same
Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	Nil

From my partition table in TableC2
Partition 1 no of sectors = 204800 sector size = 512
Total disk size = total sectors * sector size
1024bytes =1KB 1024KB = 1MB 1024MB = 1GB
To get values in KB, MB and GB divide the respective values by 1024

Item	Formula/Value	Answer
Total disk size in bytes	204,800 * 512	104,857,600 bytes
Kilobytes	104,857,600/ 1024	102,400 KB
Megabytes	102,400 / 1024	100 MB
Gigabytes	100 / 1024	0.1 GB

TableG Step-by-Step Calculation of Partition 1
From my partition table in TableC2
Partition 2 no of sectors = 32768 sector size = 512
Total disk size = total sectors * sector size
1024bytes =1KB 1024KB = 1MB 1024MB = 1GB
To get values in KB, MB and GB divide the respective values by 1024

Item	Formula/Value	Answer
Total disk size in bytes	32,768 * 512	16,777,216 bytes
Kilobytes	16,777,216 / 1024	16,384 KB
Megabytes	16,384 / 1024	16 MB
Gigabytes	16 / 1024	0.02 GB

TableG1 Step-by-Step Calculation of Partition 2

